



## MGMT 216 - Business Data Analytics -- Project (I)--

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### J&J Chocolates Online

**B**uying chocolate online is always an original way to surprise family and friends on Christmas, birthdays, Valentine's Day or other special occasions. The wide range of exquisite gift ideas you can find on their website will exceed your expectations.



*J&J Chocolates* manufactures and sells high-quality and specialty chocolate products at its factory plant and retail store located in central New Jersey. A couple of years ago, the company developed its e-commerce website and began selling its products over the Internet. Website sales have exceeded the company's expectations, and management is now considering strategy to increase sales even further. To learn more about the website customers, a sample of 50 chocolate transactions was selected from the previous month's sales. Data showing the day of the week each transaction was made, the type of payment method the customer used, the time spent on the website, the number of website pages viewed, and the amount spent by each of the 50 customers are contained in the company Excel file named *Shoppers.exl*.



J&J Chocolates would like to use the sample data to study its customers' behavior and determine if online shoppers who spend more time and view more pages also spend more money during their visit to the website. The company would also like to investigate the effect that the day of the week and the type of payment have on sales.

The project requires a report in a professional format. The total pages are limited to no more than 10 pages. You may have snapshots of Excel in the report.

### Managerial Report:

Use the methods of descriptive statistics to learn about the customers who visit the J&J Chocolates website. Include the following in your report.

1. Graphical and numerical summaries for (1) the length of time the shopper spends on the website, (2) the number of pages viewed, and (3) the mean amount spent per transaction. Discuss what you learn about J&J Chocolate's online shoppers from these numerical summaries.  
(1) Single-quantitative; (2) Single-quantitative; (3) Single-quantitative.
2. Summarize the frequency, the total dollars spent, and the mean amount spent per transaction for each day of week. What observations can you make about J&J Chocolates' business based on the day of the week, from Monday through Sunday? Please try to provide insightful discussion.  
(1) Single-categorical; (2) & (3) Cross-variable\_Crosstabulation: Categorical (Day) vs. Quantitative (Amount Spent).
3. Summarize the frequency, the total dollars spent, and the mean amount spent per transaction for each payment type. What observations can you make about J&J Chocolates' business based on the payment type? Please try to provide insightful discussion.  
(1) Single-categorical; (2) & (3) Cross-variable\_Crosstabulation: Categorical vs. Quantitative.
4. Develop a scatter diagram and compute the sample correlation coefficient to explore the relationship between (1) the time spent on the website and (2) the dollar amount spent. Use the horizontal axis for the time spent on the website. Some discussion is expected.  
Cross-variable\_Correlation Analysis: Time Spent vs. Amount Spent
5. Develop a scatter diagram and compute the sample correlation coefficient to explore the relationship between (1) the number of website pages viewed and (2) the dollar amount spent. Use the horizontal axis for the number of website pages viewed. Discuss please.  
Cross-variable\_Correlation Analysis: Pages Viewed vs. Amount Spent
6. Based on the observation and results above, please develop some strategy or business solutions for J&J Chocolates.